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ORIGINAL SCIENTIFIC PAPER
Acute Coronary Syndromes

Rescue percutaneous coronary intervention in the pharmaco-invasive era of ST-elevation myocardial infarction: insights from the STREAM-2 trial

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Aims

Contemporary guidelines support the use of a pharmaco-invasive (PI) strategy with immediate transfer to a percutaneous coronary intervention (PCI)-capable hospital for ST-elevation myocardial infarction when a timely primary PCI (pPCI) is unattainable. However, when reperfusion with fibrinolysis fails to occur, rescue PCI is recommended.

Methods and results

In a *pre-specified* analysis from STREAM-2, we explored patients randomized to PI treatment and compared those receiving half-dose tenecteplase and required rescue intervention to those with successful fibrinolysis undergoing scheduled angiography. To provide context for those randomized to pPCI, we also explored the relationship between sites of randomization, i.e. community hospital (CH) vs. ambulance on clinical outcomes. Resolution of ST-elevation following angiography and the composite of 30-day all-cause death, shock, heart failure, and reinfarction, as well as safety, reflected by stroke and non-intracranial bleeding, were measured. Of the 583 patients in the current study, 168 patients required rescue intervention (43.5%), 218 patients had successful fibrinolysis scheduled for angiography, and 197 were randomized to pPCI. Rescue PCI patients, compared with those undergoing scheduled angiography, had less ST resolution ≥50% (76.3 vs. 92.5%, $P < 0.001$) and worse clinical composite outcomes at 30 days (16.7 vs. 6.0%, $P < 0.001$) with a higher risk of intracranial haemorrhage (2.4 vs. 0.5%). Intermediate outcomes were observed for patients undergoing pPCI (ST resolution ≥50%: 78.7%; a 30-day composite outcome: 12.2%). Rescue intervention deployed in CH patients required 10 min longer compared with ambulance patients; however, there was a similar ST resolution of ≥50% (72.2 vs. 80.5%, $P = 0.219$) and comparable 30-day composite outcomes [17.6 vs. 15.7%, relative risk (RR) 0.97, 95% confidence interval (CI) 0.50–1.87], irrespective of location. Primary PCI required 48 min longer in CH patients, but resulted in similar outcomes to ambulance patients (ST resolution ≥50%: 77.0 vs. 80.2%, $P = 0.595$; 30-day composite outcome: 9.3 vs. 15.6%, RR 1.57, 95% CI 0.72–3.41, respectively).

Conclusion

Contemporary PI with half-dose tenecteplase in older patients requiring rescue intervention led to less ST resolution and worse 30-day outcomes compared with those with successful fibrinolysis receiving scheduled angiography. Notably, delays to deploying rescue PCI in CH patients were shortened over those previously achieved thereby resulting in similar outcomes to those randomized in the ambulance. Our results reinforce the benefits of functional hub and spoke models with rapid transfer to a PCI-capable facility.

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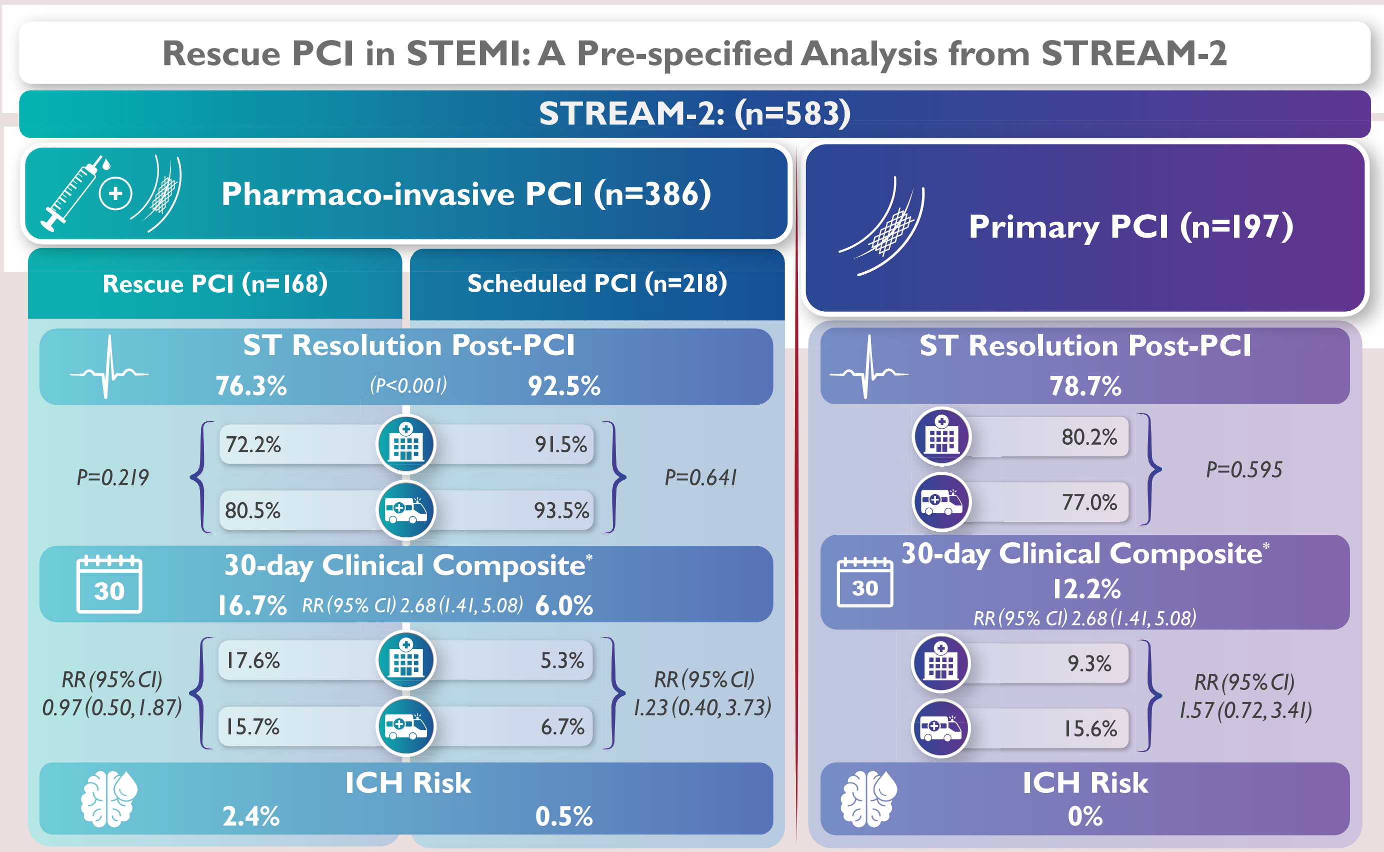
Graphical Abstract



Rescue PCI in STEMI: A pre-specified analysis from STREAM-2. *Clinical composite is all-cause death, shock, heart failure, and reinfarction. STREAM, Strategic Reperfusion Early After Myocardial Infarction; PCI, percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction; ICH, intracranial haemorrhage; RR, relative risk; CI, confidence interval.

Keywords Pharmaco-invasive • Percutaneous coronary intervention • STEMI • Older patients • Community hospital • Ambulance

Introduction

When delays to primary percutaneous coronary intervention (pPCI) are unavoidable, a pharmaco-invasive (PI) strategy with tenecteplase followed by timely PCI provides an attractive alternative reperfusion strategy in ST-elevation myocardial infarction (STEMI).[1,2] Given that reperfusion with fibrinolysis is not always successful, there is a compelling need for timely rescue coronary intervention. The REACT (Randomized Evaluation of Routine Follow-up Coronary Angiography After Percutaneous Coronary Intervention) trial demonstrated that the provision of immediate coronary angiography and rescue PCI in patients whose fibrinolysis-based reperfusion failed (<50% ST-segment resolution within 90 min) improved event-free survival and long-term mortality when compared with repeated fibrinolysis or conservative management.[3,4] International guidelines for the management of acute coronary syndromes recommend rescue PCI for failed fibrinolysis (lack of adequate ST-segment resolution) and/or the presence of electrical or haemodynamic instability or worsening ischaemia as a Class I recommendation.[1,2]

In the STREAM-1 (Strategic Reperfusion Early after Myocardial Infarction) trial, which enrolled STEMI patients unable to achieve PCI within 1 h of presentation, those randomized to a PI strategy vs. pPCI experienced similar rates of death, shock, heart failure, or reinfarction at 30 days.[5] However, patients receiving rescue PCI had worse 30-day outcomes compared with those with successful fibrinolysis receiving scheduled angiography or to those undergoing pPCI.[6] Moreover, patients randomized in a community hospital (CH) experienced longer times to rescue PCI and had worse outcomes than those in the pre-hospital ambulance setting, thereby underscoring the importance of temporal delays.[7]

More recently, in older STEMI patients ≥60 years of age who were part of STREAM-2, we found that those randomized to half-dose PI treatment showed no difference in clinical outcomes but had better ST resolution post-angiography as compared with pPCI.[8] We have also shown data that support half-dose PI treatment to be as efficacious as a full-dose PI strategy with a low systemic bleeding risk.[9,10] Unlike STREAM-1, where CHs represented only 19.2% of the STEMI population, over half of patients in STREAM-2 were randomized at CH sites, thereby providing a more robust comparison to those enrolled in ambulances. This is of particular interest given our use of half the usual dose of tenecteplase, the increased global uptake of PI therapy in patients requiring transfer to a CH,[11,12] and the persisting delays to pPCI incumbent on transfer from CH to pPCI-capable sites.[13]

Hence, in this pre-specified analysis, we examined the clinical and electrocardiographic characteristics as well as 30-day outcomes and safety in patients undergoing rescue PCI and compared them to those with successful fibrinolysis receiving scheduled angiography. Additionally, we explored the impact of commencing fibrinolytic therapy in a CH vs. ambulance location on times to rescue intervention and clinical outcomes. For both objectives, comparisons were made for those patients randomized to pPCI.

Methods

In brief, 604 patients from STREAM-2 with STEMI ≥60 years presenting within 3 h of symptoms and unable to undergo pPCI within 1 h were randomly assigned (2:1) to receive half-dose tenecteplase followed by coronary angiography—with rescue intervention if needed—otherwise 6–24 h following randomization (PI strategy), or to pPCI. Patients assigned to the PI strategy received half-dose weight-adjusted tenecteplase, 150–325 mg of aspirin, 300 mg of clopidogrel, and 0.75 mg/kg of subcutaneous enoxaparin, and an additional intravenous dose of 30 mg if <75 years of age (after amendment). Patients assigned to pPCI received the standard of care as defined by local practice. The 30-day primary clinical endpoint was all-cause death, shock, heart failure, and reinfarction; key safety criteria were total stroke, intracranial haemorrhage (ICH) and major (non-intracranial) bleeds. Rescue intervention was determined by the investigator according to the STREAM-2 protocol.[14] The current study is based on treated participants as per our prior STREAM-1 analysis (see Supplementary material online, *Figure S1*).[6] Specifically, patients were classified according to the treatment received, and those without the disease of interest (two patients with pericarditis) or did not undergo cardiac catheterization were excluded. Prior to receiving coronary angiography, there were 10 deaths in the scheduled angiography group, 3 in the pPCI group, and none in those who underwent rescue PCI.

STREAM-2 electrocardiograms (ECGs) underwent core laboratory electrocardiographic measurements of reperfusion by the ECG core laboratory at the Canadian VIGOUR Centre, University of Alberta; reperfusion efficacy was based on the proportion of patients with ≥50% ST-segment elevation resolution in the lead with worst ST-segment elevation and resolution of ST deviations (in percentages and in millimetres) after last angiography. Electrocardiogram measurements at baseline, post-reperfusion, and discharge were interpreted at the ECG core laboratory, blinded to 30-day clinical outcomes. The Selvester 54-criteria/32-point QRS scoring system, a validated ECG estimate of infarct size, was measured at hospital discharge along with Q waves. Electrocardiogram interpretation was performed using standardized ECG interpretation per guideline recommendations, unless otherwise specified. The Cardio Calipers version 3.3 programme (Iconico, Inc., New York, NY, USA) was used to perform electronic measurements. Inter- and intra-reader reliabilities were tested using Kappa or intra-class correlation coefficient wherever appropriate. Excellent laboratory inter-reader agreement has been previously established and contemporaneously monitored.[14] When there was uncertainty regarding ECG measurement, a consensus process between readers was used for final ECG reporting.

Statistical analysis

Patient characteristics, including demographic, medical history, and clinical indicators such as vital signs, time to randomization and treatment intervals, and ECG and angiographic measurements are summarized according to rescue, scheduled and pPCI patients, and then within each group according to the place of randomization. Categorical variables were reported as percentages and as median (25th and 75th percentiles) for continuous variables. Comparisons of participants' characteristics receiving rescue and scheduled angiography tested with χ^2 test (or Fisher's exact test, as appropriate) and Wilcoxon rank-sum test, respectively. Comparisons of characteristics in those randomized in the CH and ambulances were made within each treatment group.

The incidence of the primary composite endpoint and its components and safety endpoints were reported according to rescue/scheduled/pPCI and place of randomization. Kaplan–Meier curves were also plotted for these treatment groups, and the difference between rescue and scheduled angiography was tested by the log-rank test. Relative risk (RR) for the clinical outcomes in rescue vs. scheduled angiography was estimated by Poisson regression with robust error variance, providing the 95% confidence interval (CI), and was adjusted for TIMI (thrombolysis in myocardial infarction) risk score and time from symptom onset to randomization. Three rescue angiography patients and seven scheduled angiography patients were omitted from analyses due to a missing TIMI risk score; one of the scheduled patients expired prior to 30 days. Finally, the adjusted RR of the clinical outcomes were also estimated for patients randomized in ambulance vs. CH within each treatment group, and the interaction of treatment group and place of randomization was tested with the $P_{\text{interaction}}$ reported.

There was no imputation for missing data. All statistical analyses were conducted using SAS (Version Base 9.4; Cary, NC, USA). *P*-values for two-sided tests are provided for descriptive purposes and not corrected for multiple comparisons.

Results

The derivation of the three patient cohorts is shown in Supplementary material online, *Figure S1*. Of the 386 patients undergoing PI therapy, 168 required rescue intervention (43.5%), and, of these, 50.6% presented to a CH and 49.4% presented in the ambulance. Of note, 16 patients were excluded from the current study due to the lack of disease of interest or failure to receive cardiac catheterization. The reasons for rescue as defined by investigators were: failure to achieve ≥50% resolution of ST-elevation in 119 patients (70.8%); persisting chest pain in 31 (18%) (17 of these patients had ST-elevation resolution ≥50% at ≥60 min), 7 (4.3%) had haemodynamic/electrical instability, and 11 (6.5%) were undefined.

In Supplementary material online, *Table S1*, the baseline participant characteristics are shown according to each of the three treatment groups. Rescue patients were significantly less likely to be in Killip Class 1, had lower systolic blood pressures and a higher TIMI risk score as compared with those with successful fibrinolysis scheduled for coronary angiography. Time from symptom onset to randomization was similar across the three groups. Time to administration of tenecteplase was also similarly brief (10–11 min) in both PI cohorts whereas the time to achieve pPCI took ∼70 min longer. Rescue intervention was initiated at a median time of 143 min from randomization (or 245 min from symptom onset) whereas for those undergoing scheduled angiography this was 918 min. Treatment medications were similar between patients receiving rescue and scheduled angiography as seen in Supplementary material online, *Table S2*.

As seen in Supplementary material online, *Table S3*, 306 (53%) of patients presented to a CH and the remainder were randomized in the ambulance setting. The baseline characteristics of CH patients were overall similar to those randomized in ambulance, with the exception that CH patients were less commonly female and more likely to present in Killip Class II/III heart failure. Their key time intervals are also shown according to randomization site and reperfusion strategy (see Supplementary material online, *Table S3*). Community hospital patients had a 17 min longer delay from symptom onset to randomization compared with ambulance patients ($P = 0.002$). For PI patients, the time from randomization to tenecteplase administration was similar in the CH and ambulance settings (median 10 min). Additionally, similar rates of rescue intervention were observed in each location (43 vs. 44%, respectively). When rescue intervention was deployed, the median time from randomization to sheath insertion was delayed by ∼10 min in the CH (median 147 min) when compared with ambulance patients (median 137 min). Conversely, CH patients experienced a 48 min delay in time to pPCI commencement compared with ambulance patients (median 114 vs. 66 min). This resulted in an overall PCI time from symptom onset to sheath insertion that was 65 min longer for CH patients (median 215 min) compared with ambulance patients (median 150 min).

Electrocardiogram, angiographic and revascularization outcomes

Table 1 (upper panel) shows the core laboratory ECG metrics of reperfusion for each group. Times from symptom onset to qualifying ECG were similar in all groups. The worst-lead ST-elevation was similar in each group, but the sum ST deviation was numerically greater in patients with rescue intervention compared with those undergoing scheduled angiography. Approximately 90 min following tenecteplase administration, rescue patients exhibited significantly greater residual ST-elevation in the lead with worst ST-elevation compared with patients undergoing scheduled angiography. This was accompanied by markedly fewer rescue patients achieving ≥50% ST-elevation resolution and less resolution of the sum of ST deviation. Similarly, early after coronary angiography, the proportion of patients who achieved ≥50% ST-elevation resolution was less with rescue angiography, and there was also less resolution in the sum of ST deviation and greater sum of residual ST deviation as compared with patients receiving scheduled angiography. In pPCI patients at baseline, these ST-elevation and ST deviation measurements were similar to those in both PI groups: after angiography, the proportion of pPCI patients with ≥50% ST-elevation, as well as their resolution of ST deviation, appeared intermediate between the rescue and scheduled PI cohorts. At discharge, there were more Q waves within the distribution of the index myocardial infarction in those receiving rescue intervention, but similar QRS scores at discharge compared with scheduled angiography. The pPCI patients had a similar incidence of Q waves as the rescue patients and nominally higher QRS scores at discharge. Similar electrocardiographic results were demonstrated regardless of the site of randomization (Table 2, upper panel).

Table 1 (lower panel) shows the angiographic measurements and revascularization according to each group. Both pre- and post-PCI TIMI flow Grade 3 was higher in those PI patients undergoing scheduled angiography compared with rescue intervention. In pPCI patients, pre-PCI TIMI 3 flow was 18.6%; however, following PCI, TIMI 3 flow improved to 87.0% and was similar to the PI patients. Percutaneous coronary intervention was performed less frequently in PI patients undergoing scheduled angiography, and they also had numerically higher rates of coronary artery bypass graft (CABG) surgery during hospitalization than either PI or pPCI patients. Transradial intervention was employed in over 90% of all patients undergoing angiography irrespective of treatment assignment. Similar results were demonstrated irrespective of location of randomization (Table 2, lower panel).

Efficacy and safety outcomes

Table 3 (upper panel) shows clinical efficacy according to reperfusion strategy. Rescue patients had a significantly higher primary composite outcome as well as more all-cause deaths and cardiogenic shock compared with those with successful fibrinolysis scheduled for coronary angiography. One-year death was also significantly higher in those with rescue intervention. Patients undergoing pPCI had 30-day outcomes and 1-year mortality that were intermediate between those with rescue and scheduled PI therapy. The primary composite outcomes of all three groups are shown according to Kaplan–Meier curves in Figure 1. Those patients with successful fibrinolysis receiving scheduled angiography had significantly better clinical outcomes than those who underwent rescue intervention, which was characterized by an early reduction in clinical events (log-rank $P = 0.002$). Those treated with pPCI again showed intermediate clinical outcomes between the rescue and scheduled PI patients. As seen in Table 4, similar patterns were evident in the outcomes according to reperfusion strategy in both CH and ambulance patients. There was also no interaction between the location of randomization and reperfusion strategy on clinical outcomes.

The safety endpoints are shown in Tables 3 and 4 (lower panels). Four cases of ICH (three of which fatal) were observed with rescue intervention; three of these were observed in ambulance patients and one in the CH cohort. One additional ICH occurred in a PI patient randomized in the ambulance undergoing scheduled angiography (alive at 1 year). There were no ICHs observed in the pPCI group. Major non-intracranial bleeding and blood transfusions were uncommon and similar in all three groups.

Comparison to STREAM-1

To provide a historical context for interpreting the current study, we explored the relationship between the location of randomization on the times to reperfusion in the similarly designed STREAM-1 trial and STREAM-2. As shown in Supplementary material online, Table S4, the times for symptom onset to randomization were overall similar in both trials, with consistently longer times in CH vs. ambulance patients. Although there was prompt administration of tenecteplase in both studies—irrespective of randomization location—the time delay to perform rescue in CH patients in STREAM-2 was abbreviated from 179 to 147 min with a commensurate reduction of overall ischaemic times from symptom onset to sheath insertion of ∼32 min (from 314 to 253 min). The median times from randomization to pPCI for CH patients were similar in STREAM-1 and -2 (i.e. 120 and 114 min, respectively).

Discussion

In this pre-specified analysis of older STEMI patients in the STREAM-2 trial presenting within 3 h of symptom onset and randomized to a half-dose tenecteplase as part of a PI strategy, rescue PCI was undertaken in about two-fifths of patients. Following their angiographic procedure, this cohort demonstrated less ST resolution, lower TIMI 3 flows, more Q waves on discharge ECGs and worse clinical outcomes compared with those with successful fibrinolysis undergoing scheduled angiography a median of 15.3 h after randomization. Whereas the substantial proportion of our patients from CH had slightly longer total ischaemic times to rescue intervention, their site of randomization did not affect their ST resolution, angiographic results or clinical outcomes. We contend this reflects well-performing hub-spoke interfaces in the context of our contemporary trial.

Timely pPCI was achieved in the comparator randomized group within an overall median time of 83 min; however, as expected, those patients randomized at CH sites experienced a significant additional delay of 48 min: median times of 114 vs. 66 min for those transferred from CH sites vs. ambulance patients. Irrespective of this delay, patients undergoing pPCI had clinical outcomes intermediate between the two PI cohorts; moreover, those from CHs had outcomes as good as, or nominally better than, patients randomized in the ambulance, perhaps reflecting the overall achievement of timely reperfusion at the participating community sites in our trial. Importantly, the median times to pPCI in our study fall well within the guideline-recommended 120 min threshold for transfer patients in the most recent European Society of Cardiology and American College of Cardiology/American Heart Association (AHA) guidelines.[1,2] By contrast, the realities in many registries, including recent data from the AHA Mission: Lifeline and Get With The Guidelines—Coronary Artery Disease registry, reveal that only 17% of STEMI patients are treated within 120 min from arrival at a CH site and these time delays are associated with greater in-hospital mortality.[13]

Notably, among patients receiving rescue PCI in the current study, 17 presented with chest pain perceived as ischaemic despite adequate ST resolution. Furthermore, in an additional 15 cases, investigators-

Table 1 Electrocardiogram endpoints, angiographic measurements and revascularization according to rescue, scheduled and primary percutaneous coronary intervention

n	Rescue angiography 168	Scheduled angiography 218	*P*-value	pPCI 197
Electrocardiogram				
Baseline ECG				
No. of patients with baseline ECG	168 (100.0)	218 (100.0)		197 (100.0)
Time symptom onset to qualifying ECG, min	74 (51, 108)	78 (55, 113)	0.367	76 (46, 115)
Worst-lead ST-elevation, mm	3 (2, 5)	3 (2, 4)	0.230	3 (2, 5)
Sum ST deviations, mm	16 (11, 23)	14 (10, 20)	0.094	16 (11, 22)
After bolus tenecteplase				
No. of patients with paired baseline ECGs	158 (94.0)	213 (97.7)		
Time symptom onset to post-tenecteplase ECG, min	189 (164, 232)	196 (167, 227)	0.307	
Worst-lead ST-elevation, mm	2 (1, 4)	1 (0, 2)	<0.001	
Worst-lead ST-elevation resolution[a] ≥50%, no./total no. (%)	55/155 (35.5)	203/213 (95.3)	<0.001	
Sum of residual ST deviations, mm	12 (7, 20)	5 (3, 8)	<0.001	
Sum ST deviation resolution[a] (continuous), %	22 (−13, 48)	69 (48, 83)	<0.001	
After last angiography (if no PCI performed) or PCI				
No. of patients with paired baseline ECGs	162 (96.4)	203 (93.1)		191 (97.0)
Time symptom onset to post-angiography or PCI ECG, min	314 (271, 369)	1094 (662, 1401)	<0.001	263 (208, 313)
Worst-lead ST-elevation, mm	1 (1, 2)	1 (0, 1)	<0.001	1 (0, 2)
Worst-lead ST-elevation resolution[a] ≥50%, no./total no. (%)	119/156 (76.3)	186/201 (92.5)	<0.001	148/188 (78.7)
Sum of residual ST deviations, mm	7 (4, 11)	4 (2, 6)	<0.001	6 (3, 11)
Sum ST deviation resolution[a] (continuous), %	60 (29, 75)	78 (58, 89)	<0.001	62 (34, 82)
Discharge ECG				
No. of patients with discharge ECGs	146 (86.9)	193 (88.5)		172 (87.3)
Q wave at discharge	77/146 (52.7)	74/193 (38.3)	0.008	87/172 (50.6)
QRS score at discharge	3 (1, 5)	3 (1, 6)	0.715	4 (1, 7)
Days from baseline ECG to discharge ECG, day	3 (3, 4)	3 (2, 3)	0.025	3 (2, 4)
Angiographic				
TIMI blood flow				
Before PCI			<0.001	
0	68/163 (41.7)	13/210 (6.2)		110/188 (58.5)
1	14/163 (8.6)	13/210 (6.2)		13/188 (6.9)
2	34/163 (20.9)	30/210 (14.3)		30/188 (16.0)
3	47/163 (28.8)	154/210 (73.3)		35/188 (18.6)
After PCI			0.010	
0	4/165 (2.4)	3/210 (1.4)		4/193 (2.1)
1	3/165 (1.8)	2/210 (1.0)		0/193 (0.0)
2	22/165 (13.3)	9/210 (4.3)		21/193 (10.9)
3	136/165 (82.4)	196/210 (93.3)		168/193 (87.0)
Revascularization				
Radial artery access—no./total no. (%)	152/166 (91.6)	209/218 (95.9)	0.078	180/197 (91.4)
PCI—no./total no. (%)	157/168 (93.5)	187/218 (85.8)	0.016	186/197 (94.4)
Stents placement—no./total no. (%)	150/156 (96.2)	184/187 (98.4)	0.196	178/186 (95.7)
Index hospitalization CABG—no./total no. (%)	2/168 (1.2)	8/218 (3.7)	0.129	3/197 (1.5)

Continuous variables reported as median (25th and 75th percentile).
pPCI, primary percutaneous coronary intervention; ECG, electrocardiogram; TIMI, thrombolysis in myocardial infarction; CABG, coronary artery bypass graft.
[a]ST-segment resolution relative to the baseline ECG.

Table 2 Electrocardiogram endpoints, angiographic measurements and revascularization according to reperfusion strategy and place of randomization

	All patients			Rescue angiography			Scheduled angiography			pPCI		
n	Amb 277	CH 306	*P*-value	Amb 83	CH 85	*P*-value	Amb 104	CH 114	*P*-value	Amb 90	CH 107	*P*-value
Electrocardiogram												
Baseline ECG												
No. of patients with baseline ECG	277 (100.0)	306 (100.0)		83 (100.0)	85 (100.0)		104 (100.0)	114 (100.0)		90 (100.0)	107 (100.0)	
Time symptom onset to qualifying ECG, min	70 (43, 105)	81 (60, 119)	0.001	72 (46, 100)	78 (58, 116)	0.375	73 (48, 109)	85 (60, 122)	0.004	63 (38, 109)	85 (53, 119)	0.030
Worst-lead ST-elevation, mm	3 (2, 4)	3 (2, 5)	0.113	3 (2, 4)	3 (2, 5)	0.371	3 (2, 4)	3 (2, 5)	0.306	3 (2, 4)	3 (2, 5)	0.388
Sum ST deviations, mm	15 (10, 21)	16 (11, 22)	0.204	16 (11, 22)	16 (11, 23)	0.632	14 (10, 18)	14 (10, 21)	0.727	15 (11, 22)	16 (12, 22)	0.189
After bolus tenecteplase												
No. of patients with paired baseline ECGs	175 (63.2)	196 (64.1)		74 (89.2)	84 (98.8)		101 (97.1)	112 (98.2)				
Time symptom onset to post-tenecteplase ECG, min	184 (156, 215)	200 (177, 240)	<0.001	180 (153, 211)	199 (175, 244)	0.008	187 (161, 219)	200 (179, 235)	0.010			
Worst-lead ST-elevation, mm	1 (0, 2)	2 (1, 3)	0.014	2 (1, 3)	3 (2, 5)	0.052	1 (0, 1)	1 (0, 2)	0.073			
Worst-lead ST-elevation resolution[a] ≥50%, no./total no. (%)	127/173 (73.4)	131/195 (67.2)	0.193	30/72 (41.7)	25/83 (30.1)	0.134	97/101 (96.0)	106/112 (94.6)	0.630			
Sum of residual ST deviations, mm	6 (4, 12)	8 (4, 13)	0.186	11 (6, 19)	12 (8, 21)	0.153	4 (2, 7)	5 (3, 9)	0.554			
Sum ST deviation resolution[a] (continuous), %	56 (24, 75)	47 (16, 75)	0.408	29 (−10, 51)	18 (−13, 43)	0.226	68 (52, 83)	69 (43, 83)	0.968			
After last angiography (if no PCI performed) or PCI												
No. of patients with paired baseline ECGs	258 (93.1)	298 (97.4)		79 (95.2)	83 (97.6)		92 (88.5)	111 (97.4)		87 (96.7)	104 (97.2)	
Time symptom onset to post-angiography or PCI ECG, min	335 (249, 885)	360 (273, 861)	0.127	307 (265, 347)	320 (273, 375)	0.255	1057 (635, 1486)	1135 (735, 1323)	0.830	230 (180, 293)	274 (225, 321)	0.002
Worst-lead ST-elevation, mm	1 (0, 2)	1 (0, 2)	0.018	1 (0, 2)	2 (1, 2)	0.059	1 (0, 1)	1 (0, 1)	0.035	1 (0, 2)	1 (0, 2)	0.560
Worst-lead ST-resolution[a] ≥50%, no./total no. (%)	215/256 (84.0)	238/289 (82.4)	0.612	62/77 (80.5)	57/79 (72.2)	0.219	86/92 (93.5)	100/109 (91.7)	0.641	67/87 (77.0)	81/101 (80.2)	0.595
Sum of residual ST deviations, mm	5 (2, 9)	5 (3, 9)	0.500	7 (4, 11)	6 (4, 11)	0.607	3 (2, 7)	4 (3, 6)	0.207	6 (3, 11)	6 (3, 10)	0.841
Sum ST deviation resolution[a] (continuous), %	68 (40, 86)	68 (44, 83)	1.000	62 (29, 78)	58 (27, 75)	0.765	80 (58, 91)	75 (58, 88)	0.236	60 (20, 81)	64 (38, 83)	0.273
Discharge ECG												
No. of patients with discharge ECGs	228 (82.3)	283 (92.5)		72 (86.7)	74 (87.1)		83 (79.8)	110 (96.5)		73 (81.1)	99 (92.5)	
Q wave at discharge	97/228 (42.5)	141/283 (49.8)	0.101	36/72 (50.0)	41/74 (55.4)	0.513	28/83 (33.7)	46/110 (41.8)	0.253	33/73 (45.2)	54/99 (54.5)	0.226

Continued

Table 2 Continued

	All patients			Rescue angiography			Scheduled angiography			pPCI		
	Amb	CH	*P*-value	Amb	CH	*P*-value	Amb	CH	*P*-value	Amb	CH	*P*-value
n	277	306		83	85		104	114		90	107	
QRS score at discharge	3 (1, 5)	3 (1, 7)	0.117	3 (1, 5)	3 (1, 6)	0.853	3 (1, 5)	4 (1, 6)	0.176	4 (1, 6)	4 (2, 7)	0.189
Days from baseline ECG to discharge ECG, day	3 (3, 4)	3 (2, 3)	<0.001	3 (3, 4)	3 (2, 4)	0.540	3 (3, 4)	3 (2, 3)	<0.001	3 (3, 4)	3 (2, 3)	0.014
Angiographic												
TIMI blood flow												
Before PCI:			0.216			0.279			0.101			0.721
0	91/271 (33.6)	100/290 (34.5)		33/83 (39.8)	35/80 (43.8)		6/99 (6.1)	7/111 (6.3)		52/89 (58.4)	58/99 (58.6)	
1	20/271 (7.4)	20/290 (6.9)		5/83 (6.0)	9/80 (11.3)		9/99 (9.1)	4/111 (3.6)		6/89 (6.7)	7/99 (7.1)	
2	37/271 (13.7)	57/290 (19.7)		16/83 (19.3)	18/80 (22.5)		9/99 (9.1)	21/111 (18.9)		12/89 (13.5)	18/99 (18.2)	
3	123/271 (45.4)	113/290 (39.0)		29/83 (34.9)	18/80 (22.5)		75/99 (75.8)	79/111 (71.2)		19/89 (21.3)	16/99 (16.2)	
After PCI:			0.332			0.265			0.229			0.021
0	8/270 (3.0)	3/298 (1.0)		2/83 (2.4)	2/82 (2.4)		2/99 (2.0)	1/111 (0.9)		4/88 (4.5)	0/105 (0.0)	
1	3/270 (1.1)	2/298 (0.7)		1/83 (1.2)	2/82 (2.4)		2/99 (2.0)	0/111 (0.0)		0/88 (0.0)	0/105 (0.0)	
2	26/269 (9.6)	26/298 (8.7)		7/83 (8.4)	15/82 (18.3)		6/99 (6.1)	3/111 (2.7)		13/88 (14.8)	8/105 (7.6)	
3	233/270 (86.3)	267/298 (89.6)		73/83 (88.0)	63/82 (76.8)		89/99 (89.9)	107/111 (96.4)		71/88 (80.7)	97/105 (92.4)	
Revascularization												
Radial artery access—no./total no. (%)	258/277 (93.1)	283/304 (93.1)	0.982	76/83 (91.6)	76/83 (91.6)	1.000	97/104 (93.3)	112/114 (98.2)	0.065	85/90 (94.4)	95/107 (88.8)	0.159
PCI—no./total no. (%)	246/277 (88.8)	284/306 (92.8)	0.093	74/83 (89.2)	83/85 (97.6)	0.026	87/104 (83.7)	100/114 (87.7)	0.391	85/90 (94.4)	101/107 (94.4)	0.987
Stents placement—no./total no. (%)	240/246 (97.6)	272/283 (96.1)	0.346	74/74 (100.0)	76/82 (92.7)	0.018	85/87 (97.7)	99/100 (99.0)	0.481	81/85 (95.3)	97/101 (96.0)	0.803
Index hospitalization CABG—no./total no. (%)	6/277 (2.2)	7/306 (2.3)	0.921	1/83 (1.2)	1/85 (1.2)	0.987	4/104 (3.8)	4/114 (3.5)	0.895	1/90 (1.1)	2/107 (1.9)	0.665

Continuous variables reported as median (25th and 75th percentile).
Amb, ambulance; CH, community hospital; pPCI, primary percutaneous coronary intervention; ECG, electrocardiogram; TIMI, thrombolysis in myocardial infarction; CABG, coronary artery bypass graft.
[a]ST-segment resolution relative to the baseline ECG.

Table 3 Clinical efficacy and safety endpoints according to rescue, scheduled and primary percutaneous coronary intervention

	Rescue angiography	Scheduled angiography	Rescue vs. scheduled	pPCI
Clinical efficacy endpoints	**Observed event rate (%)**		**RR (95% CI)[a]**	
N	**168**	**218**		**197**
Composite clinical efficacy endpoint: all-cause death, heart failure, shock, reinfarction at 30 days, *n* (%)	28 (16.7)	13 (6.0)	2.65 (1.40, 5.03)	24 (12.2)
Death from any cause	21 (12.6)	6 (2.8)	4.85 (1.86, 12.6)	15 (7.6)
Cardiogenic shock	12 (7.2)	1 (0.5)	—	9 (4.6)
Heart failure	8 (4.8)	5 (2.3)	1.77 (0.58, 5.40)	9 (4.6)
Reinfarction	4 (2.4)	6 (2.8)	0.89 (0.25, 3.19)	5 (2.5)
Death from any cause at 1 year, *n* (%)	27 (16.1)	9 (4.1)	4.39 (1.95, 9.88)	22 (11.2)
Safety endpoints, *n* (%)				
Total stroke	5 (3.0)	3 (1.4)	2.39 (0.60, 9.56)	1 (0.5)
ICH	4 (2.4)	1 (0.5)	—	0 (0.0)
Ischaemic stroke	1 (0.6)	2 (0.9)	—	1 (0.5)
Major non-intracranial bleeding	3 (1.8)	2 (0.9)	1.95 (0.35, 10.9)	2 (1.0)
Blood transfusion	1 (0.6)	1 (0.5)	—	1 (0.5)

pPCI, primary percutaneous coronary intervention; RR, relative risk; CI, confidence interval; ICH, intracranial haemorrhage; TIMI, thrombolysis in myocardial infarction.
[a]Relative risk adjusted by TIMI risk score and time from symptom onset to randomization. Relative risk not reported for events less and equal to 1 in any cell.

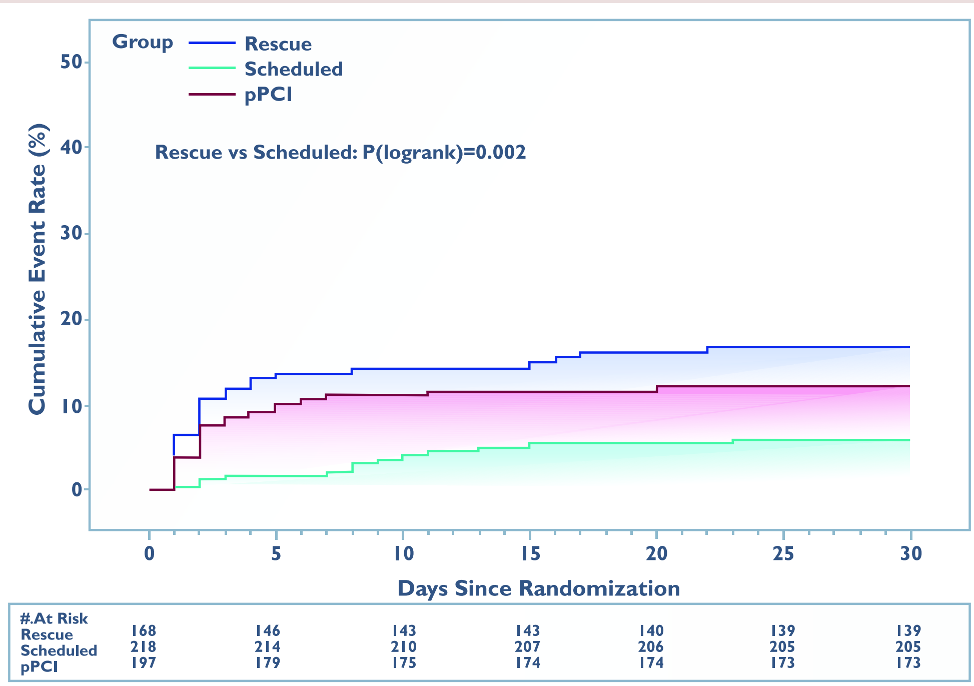


Figure 1 Kaplan–Meier curves depicting the 30-day primary composite efficacy outcome of death, shock, heart failure and reinfarction for the rescue, scheduled and primary percutaneous coronary intervention groups. pPCI, primary percutaneous coronary intervention.

initiated rescue PCI *prior to the protocol-mandated ECG assessment* at 60–90 min post-fibrinolytic therapy. It is noteworthy that those patients selected for rescue were less likely to be in Killip Class 1, have lower systolic blood pressures and higher TIMI risk scores. Although half-dose of tenecteplase was used in our study, we observed five ICH cases. Four of these were associated with rescue intervention

Table 4 Clinical efficacy and safety endpoints at 30 days according to reperfusion strategy and place of randomization

	All patients			Rescue angiography			Scheduled angiography			pPCI			
	Amb	CH	RR (95% CI)[a]	Amb	CH	RR (95% CI)[a]	Amb	CH	RR (95% CI)[a]	Amb	CH	RR (95% CI)[a]	$P_{interaction}$
N	277	306		83	85		104	114		90	107		
Clinical efficacy endpoints													
Composite clinical efficacy endpoint: all-cause death, heart failure, shock, reinfarction at 30 days	34 (12.3)	31 (10.1)	1.22 (0.77, 1.95)	13 (15.7)	15 (17.6)	0.97 (0.50, 1.87)	7 (6.7)	6 (5.3)	1.23 (0.40, 3.73)	14 (15.6)	10 (9.3)	1.57 (0.72, 3.41)	0.712
Death from any cause	25 (9.0)	17 (5.6)	1.60 (0.87, 2.93)	11 (13.3)	10 (11.9)	1.23 (0.56, 2.71)	5 (4.8)	1 (0.9)	5.08 (0.54, 48.0)	9 (10.0)	6 (5.6)	1.54 (0.55, 4.30)	0.365
Cardiogenic shock	9 (3.2)	13 (4.3)	0.83 (0.33, 2.05)	4 (4.8)	8 (9.5)	NA	0 (0.0)	1 (0.9)	NA	5 (5.6)	4 (3.7)	NA	NA
Heart failure	8 (2.9)	14 (4.6)	0.63 (0.26, 1.53)	2 (2.4)	6 (7.1)	0.38 (0.08, 1.80)	2 (1.9)	3 (2.6)	0.84 (0.14, 4.99)	4 (4.4)	5 (4.7)	0.81 (0.20, 3.27)	0.598
Reinfarction	9 (3.2)	6 (2.0)	1.50 (0.51, 4.40)	2 (2.4)	2 (2.4)	1.03 (0.15, 7.28)	3 (2.9)	3 (2.6)	1.11 (0.22, 5.49)	4 (4.4)	1 (0.9)	3.72 (0.39, 35.1)	0.715
Death from any cause at 1-year, *n* (%)	29 (10.5)	29 (9.5)	1.14 (0.69, 1.88)	12 (14.5)	15 (17.6)	0.89 (0.45, 1.77)	7 (6.7)	2 (1.8)	7.50 (0.87, 64.7)	10 (11.1)	12 (11.2)	0.87 (0.38, 2.02)	0.218
Safety endpoints													
Total stroke	5 (1.8)	4 (1.3)		4 (4.8)	1 (1.2)		1 (1.0)	2 (1.8)		0 (0.0)	1 (0.9)		
ICH	4 (1.4)	1 (0.3)		3 (3.6)	1 (1.2)		1 (1.0)	0 (0.0)		0 (0.0)	0 (0.0)		
Ischaemic stroke	1 (0.4)	3 (1.0)		1 (1.2)	0 (0.0)		0 (0.0)	2 (1.8)		0 (0.0)	1 (0.9)		
Major non-intracranial bleeding	5 (1.8)	2 (0.7)		2 (2.4)	1 (1.2)		2 (1.9)	0 (0.0)		1 (1.1)	1 (0.9)		
Blood transfusion	2 (0.7)	1 (0.3)		1 (1.2)	0 (0.0)		1 (1.0)	0 (0.0)		0 (0.0)	1 (0.9)		

Amb, ambulance; CH, community hospital; pPCI, primary percutaneous coronary intervention; PI, pharmaco-invasive; RR, relative risk; CI, confidence interval; ICH, intracranial haemorrhage; n/a, not applicable.
[a]Relative risk and 95% confident intervals for comparison between pre-hospital and community hospital, and adjusted for TIMI risk score and time from symptom onset to randomization.

(three in the ambulance cohort and fatal), while only one occurred in a patient with scheduled angiography (alive at 1 year). The urgency associated with transfer of patients for rescue procedures was associated with excess (crossover) anticoagulation[10] accentuated in part by communication challenges between different healthcare teams characteristic of the fast-paced rescue setting. This underscores the critical importance of allowing adequate time for pharmacologic therapy to work and only proceeding with rescue when absolutely indicated. The incidence of major non-intracranial bleeding and blood transfusion was low, probably attributable to the high use of radial vascular access.

Given the continuing role of CHs in the global care of STEMI patients and our prior findings of the negative consequences of rescue coronary intervention for patients in CH sites in STREAM-1, it is interesting to note the different outcomes of rescue PCI in the STREAM-1 and -2 studies. In STREAM-1, the 30-day primary composite event rate for rescue patients randomized in CH was 30.9% with full-dose tenecteplase vs. 18.2% for ambulance patients.[6] By contrast, in STREAM-2, the overall 30-day composite event rate for rescue intervention was 16.7% and CH patients fared similarly well to ambulance patients (17.6 vs. 15.9%, respectively). Although the times to perform rescue angiography between the two studies were similar, the delay for CH patients was markedly shorter in STREAM-2 such that rescue PCI was performed 32 min faster. Moreover, we found similar post-angiographic ST-segment resolution (and TIMI flow) resulting in at least as good 30-day primary outcomes with rescue PCI from the CH compared with ambulance patients. Mitigating delays highlight the importance of expeditious transfer of these rescue patients from a CH and reinforces the importance of a hub and spoke model to STEMI care, where delays to transfer can be minimized in the community with rapid ambulance transport immediately following the administration of fibrinolysis to the PCI-capable hospital where determination of reperfusion can be assessed for the need of rescue intervention.[15,16]

The optimal timing for the performance of angiography and PCI after fibrinolysis is uncertain and recent guidelines from both Europe and the American College Cardiology/American Heart Association indicate 'it has not been clearly defined' and has a wide range up to 24 h.[1,2] It is interesting to note that here was a nominally less performance of PCI and a doubling of CABG performed in-hospital in those with scheduled angiography compared with those with rescue intervention (or even pPCI). Perhaps, as noted previously, this allows for a more contemplative approach to address both the completeness of thrombus resolution over time. Moreover, a more nuanced approach in those with complex multi-vessel disease not requiring emergent PCI may be useful in some patients in whom surgical revascularization may be a useful alternative.[5]

Study limitations

Our study has both strengths and limitations. Our analysis was a pre-specified comparison of reperfusion strategies with complete ascertainment of clinical endpoints captured within a randomized trial in those randomized in a CH site or in the ambulance, with independent core laboratory ECG findings. We provide novel findings on the use and outcomes of a half-dose PI strategy in older patients deemed to require rescue PCI. We also acknowledge the following limitations: (i) subjectivity associated with post-randomization deployment of rescue intervention vs. scheduled PCI may be a confounding factor; (ii) the comparisons with patients randomized to pPCI need to be similarly viewed with caution; and (iii) the interactions between setting of randomization (CH or ambulance) and reperfusion strategy are exploratory given the size limitations of the study cohort. And finally, our conclusions apply only to STEMI patients presenting within 3 h of symptom onset.

Conclusions

In a contemporary analysis of older STEMI patients in the STREAM-2 study, PI patients treated with half-dose tenecteplase requiring rescue intervention had worse 30-day outcomes compared with those with successful fibrinolysis receiving scheduled angiography. Despite temporal delays in CH patients, patients requiring rescue PCI had similar outcomes to those randomized in the ambulance setting, likely related to fast transfer times. Our results underscore the importance of a well-defined STEMI network with a hub and spoke model to facilitate rapid transfer of community patients to a PCI-capable hospital.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Data availability

After 2 years from the time of the primary publication, data requests will be considered, provided the requests are made in writing by qualified researchers.

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References

1. Byrne RA, Rossello X, Coughlan JJ, Barbato E, Berry C, Chieffo A, *et al.* 2023 ESC guidelines for the management of acute coronary syndromes: developed by the task force on the management of acute coronary syndromes of the European Society of Cardiology (ESC). *Eur Heart J* 2023;**44**:3720–3826.
2. Rao SV, O'Donoghue ML, Ruel M, Rab T, Tamis-Holland JE, Alexander JH, *et al.* 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Circulation* 2025;**151**: e771–e862.
3. Gershlick AH, Stephens-Lloyd A, Hughes S, Abrams KR, Stevens SE, Uren NG, *et al.* Rescue angioplasty after failed thrombolytic therapy for acute myocardial infarction. *N Engl J Med* 2005;**353**:2758–2768.
4. Carver A, Rafelt S, Gershlick AH, Fairbrother KL, Hughes S, Wilcox R. Longer-term follow-up of patients recruited to the REACT (Rescue Angioplasty Versus Conservative Treatment or Repeat Thrombolysis) trial. *J Am Coll Cardiol* 2009;**54**: 118–126.
5. Armstrong PW, Gershlick AH, Goldstein P, Wilcox R, Danays T, Lambert Y, *et al.* Fibrinolysis or primary PCI in ST-segment elevation myocardial infarction. *N Engl J Med* 2013;**368**:1379–1387.
6. Welsh RC, Van De Werf F, Westerhout CM, Goldstein P, Gershlick AH, Wilcox RG, *et al.* Outcomes of a pharmacoinvasive strategy for successful versus failed fibrinolysis and primary percutaneous intervention in acute myocardial infarction (from the strategic reperfusion early after myocardial infarction [STREAM] study). *Am J Cardiol* 2015;**114**:811–819.
7. Welsh RC, Goldstein P, Sinnaeve P, Ostojic MC, Zheng Y, Danays T, *et al.* Relationship between community hospital versus pre-hospital location of randomisation and clinical outcomes in ST-elevation myocardial infarction patients: insights from the Stream study. *Eur Heart J Acute Cardiovasc Care* 2018;**7**:504–513.
8. Van De Werf F, Ristić AD, Averkov OV, Arias-Mendoza A, Lambert Y, Kerr Saraiva JF, *et al.* STREAM-2: half-dose tenecteplase or primary percutaneous coronary intervention in older patients with ST-segment-elevation myocardial infarction: a randomized, open-label trial. *Circulation* 2023;**148**:753–764.
9. Bainey KR, Armstrong PW, Zheng Y, Brass N, Tyrrell BD, Leung R, *et al.* Pharmacoinvasive strategy versus primary percutaneous coronary intervention in

ST-elevation myocardial infarction in clinical practice: insights from the vital heart response registry. *Circ Cardiovasc Interv* 2019;**12**:e008059.
10. Bainey KR, Welsh RC, Zheng Y, Arias-Mendoza A, Ristić AD, Averkov OV, *et al.* Pharmaco-invasive strategy and dosing of tenecteplase in STEMI patients 60 to <75 years: an inter-trial comparison of the STREAM-1 and STREAM-2 trials. *Am Heart J* 2025;**284**:20–31.
11. Rosselló X, Huo Y, Pocock S, Van de Werf F, Chin CT, Danchin N, *et al.* Global geographical variations in ST-segment elevation myocardial infarction management and post-discharge mortality. *Int J Cardiol* 2017;**245**:27–34.
12. Araiza-Garaygordobil D, Alexander T, Huber K, Halvorsen S, Ahrens I, Alviar C, *et al.* Reperfusion therapy for ST elevation myocardial infarction in low- to middle-income countries: a clinical consensus statement of the Association for Acute CardioVascular Care (ACVC), the European Association of Percutaneous Cardiovascular Interventions (EAPCI), the European Association of Preventive Cardiology (EAPC), the ESC Working Group on Thrombosis, and the Stent - Save a Life! Initiative. *Eur Heart J Acute Cardiovasc Care* 2025;**14**:690–697.
13. Jollis JG, Granger CB, Zègre-Hemsey JK, Henry TD, Goyal A, Tamis-Holland JE, *et al.* Treatment time and in-hospital mortality among patients with ST-segment elevation myocardial infarction, 2018–2021. *JAMA* 2022;**328**:2033–2040.
14. Armstrong PW, Bogaerts K, Welsh R, Sinnaeve PR, Goldstein P, Pages A, *et al.* The second strategic reperfusion early after myocardial infarction (STREAM-2) study optimizing pharmacoinvasive reperfusion strategy in older ST-elevation myocardial infarction patients. *Am Heart J* 2020;**226**:140–146.
15. Cantor WJ, Fitchett D, Borgundvaag B, Ducas J, Heffernan M, Cohen EA, *et al.* Routine early angioplasty after fibrinolysis for acute myocardial infarction. *N Engl J Med* 2009; **360**:2705–2718.
16. Bøhmer E, Hoffmann P, Abdelnoor M, Arnesen H, Halvorsen S. Efficacy and safety of immediate angioplasty versus ischemia-guided management after thrombolysis in acute myocardial infarction in areas with very long transfer distances results of the NORDISTEMI (NORwegian study on DIstrict treatment of ST-elevation my. *J Am Coll Cardiol* 2010;**55**:102–110.